

2024 AMC 10B Problem 13 - Solution

Review the full statement and step-by-step solution for 2024 AMC 10B Problem 13. Great practice for AMC 10, AMC 12, AIME, and other math contests

2024 AMC 10B Problem 13

**Problem:**

Positive integers x and y satisfy the equation $\sqrt{x} + \sqrt{y} = \sqrt{1183}$. What is the minimum possible value of $x + y$?

Answer Choices:

- A. 585
- B. 595
- C. 623
- D. 700
- E. 791

Solution:

Observe that $1183 = 13^2 \cdot 7$, so $\sqrt{1183} = 13\sqrt{7}$. Because $\sqrt{x} + \sqrt{y} = 13\sqrt{7}$, it follows that $\sqrt{x}$ and $\sqrt{y}$ must be of the form $a\sqrt{7}$ and $b\sqrt{7}$, respectively, where a and b are positive integers and $a + b = 13$. Then $\sqrt{x} = \sqrt{7a^2}$ and $\sqrt{y} = \sqrt{7b^2}$, so $x = 7a^2$ and $y = 7b^2$. Substituting gives

$$x + y = 7(a^2 + b^2) = 7(a^2 + (13 - a)^2) = 14a^2 - 182a + 1183$$

The minimum value of the quadratic polynomial occurs at $a = \frac{182}{2 \cdot 14} = 6.5$. Because a and b must be positive integers, without loss of generality (by symmetry), choose $a = 6$ and $b = 7$. The minimum possible value of $x + y$ is

$$x + y = 7 \cdot (6^2 + 7^2) = 7 \cdot (36 + 49) = 7 \cdot 85 = (\text{B}) \boxed{595}$$

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